

**Section: 1. PRODUCT AND COMPANY IDENTIFICATION**

Product name : HOROLITH FL

Other means of identification : Not applicable.

Recommended use : Surface cleaner

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 1.0 % - 2.0 %

Company : ECOLAB INTERNATIONAL SDN. BHD.  
Suite 12-01, 12-02 & 12-03A, Level 12, The Pinnacle, Persiaran  
Lagoon, Bandar Sunway  
46150 Petaling Jaya, Selangor Malaysia  
603-7628 5200

Emergency telephone number : 1800 181 880

Issuing date : 02.02.2022

**Section: 2. HAZARDS IDENTIFICATION**
**GHS Classification**
**Product AS SOLD**

Skin corrosion/irritation : Category 1A

Serious eye damage/eye irritation : Category 1

**Product AT USE DILUTION**

Not a hazardous substance or mixture.

**GHS Label element**
**Product AS SOLD**

Hazard pictograms :



Signal Word : Danger

Hazard Statements : Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**  
Do not breathe dusts or mists. Wash skin thoroughly after handling.  
Wear protective gloves/ protective clothing/ eye protection/ face protection.  
**Response:**  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/ physician. IF IN EYES: Rinse cautiously with water for several minutes. Remove

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contact lenses, if present and easy to do so. Continue rinsing. Wash contaminated clothing before reuse.

**Storage:**

Store locked up.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

### Product AT USE DILUTION

Precautionary Statements

**Prevention:**

Wash hands thoroughly after handling.

**Response:**

Get medical advice/ attention if you feel unwell.

**Storage:**

Store in accordance with local regulations.

### Product AS SOLD

Other hazards

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

## Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

### Product AS SOLD

Pure substance/mixture

: Mixture

| Chemical Name   | CAS-No.   | Concentration: (%) |
|-----------------|-----------|--------------------|
| Nitric acid     | 7697-37-2 | 30 - 60            |
| Phosphoric acid | 7664-38-2 | 1 - 5              |

### Product AT USE DILUTION

No hazardous ingredients

## Section: 4. FIRST AID MEASURES

### Product AS SOLD

In case of eye contact

: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact

: Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed

: Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled

: Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders

: If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician

: Treat symptomatically.

Most important symptoms and effects, both acute and delayed

: See Section 11 for more detailed information on health effects and symptoms.

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### Product AT USE DILUTION

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.                           |
| In case of skin contact | : Rinse with plenty of water.                           |
| If swallowed            | : Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled              | : Get medical attention if symptoms occur.              |

## Section: 5. FIREFIGHTING MEASURES

### Product AS SOLD

|                                               |                                                                                                                                                                                 |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media                  | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                                                       |
| Unsuitable extinguishing media                | : None known.                                                                                                                                                                   |
| Specific hazards during firefighting          | : Exposure to decomposition products may be a hazard to health.                                                                                                                 |
| Hazardous combustion products                 | : Decomposition products may include the following materials:<br>nitrogen oxides (NOx)<br>Oxides of phosphorus                                                                  |
| Special protective equipment for firefighters | : Use personal protective equipment.                                                                                                                                            |
| Specific extinguishing methods                | : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

## Section: 6. ACCIDENTAL RELEASE MEASURES

### Product AS SOLD

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal precautions, protective equipment and emergency procedures | : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.                                  |
| Environmental precautions                                           | : Do not allow contact with soil, surface or ground water.                                                                                                                                                                                                                                                                                                                                                    |
| Methods and materials for containment and cleaning up               | : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. |

### Product AT USE DILUTION

|                                                                     |                                                            |
|---------------------------------------------------------------------|------------------------------------------------------------|
| Personal precautions, protective equipment and emergency procedures | : Refer to protective measures listed in sections 7 and 8. |
| Environmental precautions                                           | : No special environmental precautions required.           |

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Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

### Section: 7. HANDLING AND STORAGE

#### Product AS SOLD

Advice on safe handling : Do not ingest. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 40 °C

#### Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

### Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Product AS SOLD

##### Components with workplace control parameters

| Components      | CAS-No.   | Form of exposure | Permissible concentration      | Basis  |
|-----------------|-----------|------------------|--------------------------------|--------|
| Nitric acid     | 7697-37-2 | TWA              | 2 ppm<br>5.2 mg/m <sup>3</sup> | MY OEL |
| Phosphoric acid | 7664-38-2 | TWA              | 1 mg/m <sup>3</sup>            | MY OEL |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
butyl-rubber  
Nitrile rubber  
Unsupported neoprene  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

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|                        |                                                                                                                                                                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Skin protection        | : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing                                                                                                                                                                                                       |
| Respiratory protection | : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>An acid gas cartridge may be used.                                                                                                                                                             |
| Hygiene measures       | : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard. |

#### Product AT USE DILUTION

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| Engineering measures | : Good general ventilation should be sufficient to control worker exposure to airborne contaminants. |
|----------------------|------------------------------------------------------------------------------------------------------|

#### Personal protective equipment

|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| Eye protection         | : No special protective equipment required.                       |
| Hand protection        | : No special protective equipment required.                       |
| Skin protection        | : No special protective equipment required.                       |
| Respiratory protection | : No personal respiratory protective equipment normally required. |

### Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD       | Product AT USE DILUTION |
|-----------------------------------------|-----------------------|-------------------------|
| Appearance                              | : liquid              | liquid                  |
| Colour                                  | : clear, light yellow | colourless              |
| Odour                                   | : slight              | odourless               |
| pH                                      | : 1.0, (100 %)        | 0.9 - 1.65              |
| Flash point                             | : Not applicable.     |                         |
| Odour Threshold                         | : no data available   |                         |
| Melting point/freezing point            | : no data available   |                         |
| Initial boiling point and boiling range | : > 100 °C            |                         |
| Evaporation rate                        | : no data available   |                         |
| Flammability (solid, gas)               | : Not applicable.     |                         |
| Upper explosion limit                   | : no data available   |                         |
| Lower explosion limit                   | : no data available   |                         |
| Vapour pressure                         | : no data available   |                         |
| Relative vapour density                 | : no data available   |                         |
| Relative density                        | : 1.3 - 1.34          |                         |
| Water solubility                        | : soluble             |                         |
| Solubility in other solvents            | : no data available   |                         |
| Partition coefficient: n-octanol/water  | : no data available   |                         |

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|                           |                     |
|---------------------------|---------------------|
| Auto-ignition temperature | : no data available |
| Thermal decomposition     | : no data available |
| Viscosity, kinematic      | : no data available |
| Explosive properties      | : no data available |
| Oxidizing properties      | : no data available |
| Molecular weight          | : no data available |
| VOC                       | : no data available |

#### Section: 10. STABILITY AND REACTIVITY

##### Product AS SOLD

|                                    |                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                                                                |
| Chemical stability                 | : Stable under normal conditions.                                                                                            |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas.                                            |
| Conditions to avoid                | : None known.                                                                                                                |
| Incompatible materials             | : Bases<br>Metals                                                                                                            |
| Hazardous decomposition products   | : In case of fire hazardous decomposition products may be produced such as:<br>nitrogen oxides (NOx)<br>Oxides of phosphorus |

#### Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

##### Potential Health Effects

##### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

##### Product AT USE DILUTION

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Health injuries are not known or expected under normal use. |
| Skin             | : Health injuries are not known or expected under normal use. |
| Ingestion        | : Health injuries are not known or expected under normal use. |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

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### Experience with human exposure

#### Product AS SOLD

|              |                                 |
|--------------|---------------------------------|
| Eye contact  | : Redness, Pain, Corrosion      |
| Skin contact | : Redness, Pain, Corrosion      |
| Ingestion    | : Corrosion, Abdominal pain     |
| Inhalation   | : Respiratory irritation, Cough |

#### Product AT USE DILUTION

|              |                                  |
|--------------|----------------------------------|
| Eye contact  | : No symptoms known or expected. |
| Skin contact | : No symptoms known or expected. |
| Ingestion    | : No symptoms known or expected. |
| Inhalation   | : No symptoms known or expected. |

### Toxicity

#### Product AS SOLD Product

|                                   |                     |
|-----------------------------------|---------------------|
| Acute oral toxicity               | : no data available |
| Acute inhalation toxicity         | : no data available |
| Acute dermal toxicity             | : no data available |
| Skin corrosion/irritation         | : no data available |
| Serious eye damage/eye irritation | : no data available |
| Respiratory or skin sensitization | : no data available |
| Carcinogenicity                   | : no data available |
| Reproductive effects              | : no data available |
| Germ cell mutagenicity            | : no data available |
| Teratogenicity                    | : no data available |
| STOT - single exposure            | : no data available |
| STOT - repeated exposure          | : no data available |
| Aspiration toxicity               | : no data available |

#### Components

|                     |                                              |
|---------------------|----------------------------------------------|
| Acute oral toxicity | : Phosphoric acid<br>LD50 rat: > 2,600 mg/kg |
|---------------------|----------------------------------------------|

#### Components

|                           |                                                                             |
|---------------------------|-----------------------------------------------------------------------------|
| Acute inhalation toxicity | : Phosphoric acid<br>4 h LC50 rat: 0.962 mg/l<br>Test atmosphere: dust/mist |
|---------------------------|-----------------------------------------------------------------------------|

#### Components

|                       |                                                 |
|-----------------------|-------------------------------------------------|
| Acute dermal toxicity | : Phosphoric acid<br>LD50 rabbit: > 2,000 mg/kg |
|-----------------------|-------------------------------------------------|

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### Section: 12. ECOLOGICAL INFORMATION

#### Toxicity

##### Product AS SOLD

Environmental Effects : Harmful to aquatic life.

##### Product AT USE DILUTION

Environmental Effects : This product has no known ecotoxicological effects.

##### Product AS SOLD

##### Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

#### Components

Toxicity to fish : Nitric acid  
96 h LC50: 72 mg/l

#### Components

Toxicity to daphnia and other aquatic invertebrates : Phosphoric acid  
48 h EC50 Daphnia magna (Water flea): > 100 mg/l

#### Components

Toxicity to algae : Phosphoric acid  
72 h EC50 Desmodesmus subspicatus (green algae): > 100 mg/l

#### Persistence and degradability

Not applicable - inorganic

#### Bioaccumulative potential

no data available

#### Mobility in soil

no data available

#### Other adverse effects

no data available

### Section: 13. DISPOSAL CONSIDERATIONS

##### Product AS SOLD

Disposal methods : Do not contaminate storm water drains, natural waterways or soil with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of contents/container in accordance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.



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#### Product AT USE DILUTION

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of contents/container in accordance with local regulations Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

#### Section: 14. TRANSPORT INFORMATION

##### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

##### Land transport

UN number : 2031  
Proper shipping name : NITRIC ACID  
Class : 8  
Packing group : II  
Hazchem Code : 2R  
Environmentally hazardous : No

##### Sea transport (IMDG/IMO)

UN number : 2031  
Proper shipping name : NITRIC ACID  
Class : 8  
Packing group : II  
Marine pollutant : No

#### Section: 15. REGULATORY INFORMATION

##### National Regulations

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulation 2013  
Occupational Safety and Health Act 1994  
Environmental Quality Act 1974

**The components of this product are reported in the following inventories:**

##### United States TSCA Inventory :

All substances listed as active on the TSCA inventory

##### Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL.

##### Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :

On the inventory, or in compliance with the inventory

**New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :**  
not determined

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**Japan. ENCS - Existing and New Chemical Substances Inventory :**

On the inventory, or in compliance with the inventory

**Korea. Korean Existing Chemicals Inventory (KECI) :**

On the inventory, or in compliance with the inventory

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On the inventory, or in compliance with the inventory

**China Inventory of Existing Chemical Substances :**

On the inventory, or in compliance with the inventory

**Taiwan Chemical Substance Inventory :**

On the inventory, or in compliance with the inventory

### Section: 16. OTHER INFORMATION

|                     |                      |
|---------------------|----------------------|
| Issuing date        | : 02.02.2022         |
| Date of first issue | : 17.07.2014         |
| version             | : 1.2                |
| Prepared by         | : Regulatory Affairs |

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.